

Age-Invariant Face Recognition Using Deep Learning

By: Eric Cao, Yanjun Meng, Akhil Rajavaram

Agenda



Introduction



Background



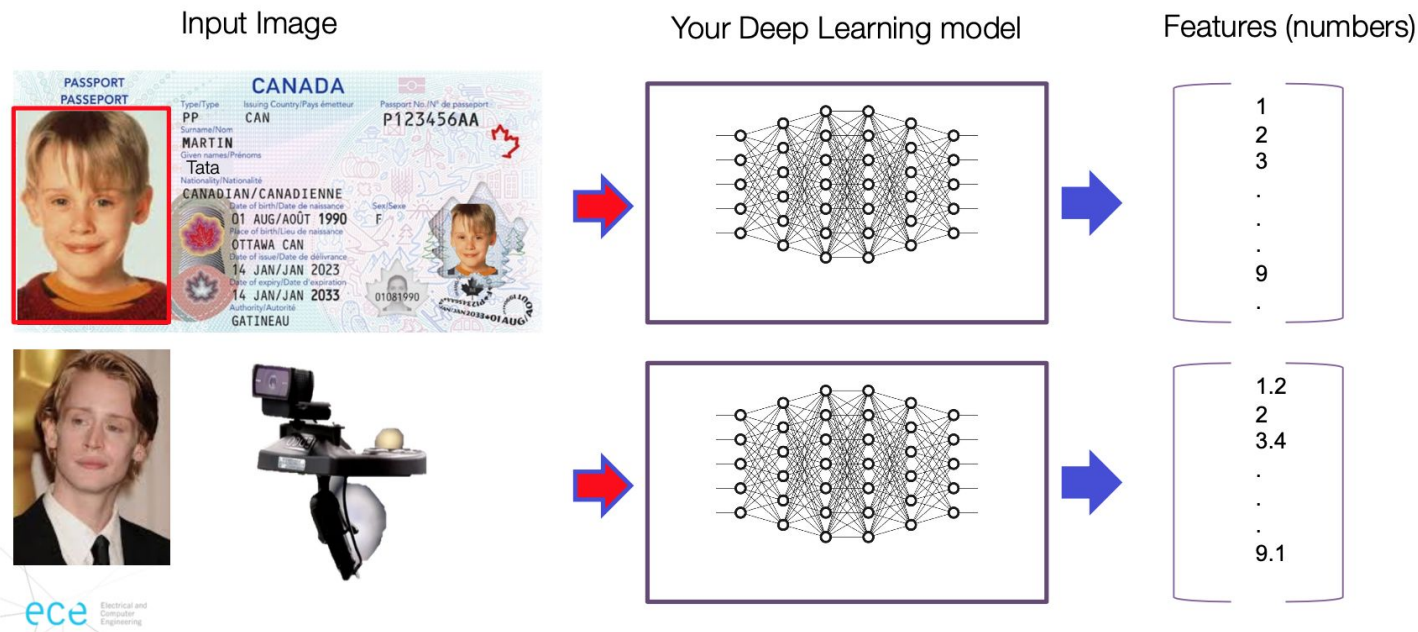
Methodology



Timeline

Introduction – Goal

- Design a deep learning network to detect identity regardless of age



Demonstration from course materials

Introduction – Applications

- Identification at airport security, border control, banks, etc.
- Finding missing persons, tracking criminals, etc.



Real Image
21 months



11 - 14 years



17 - 20 years

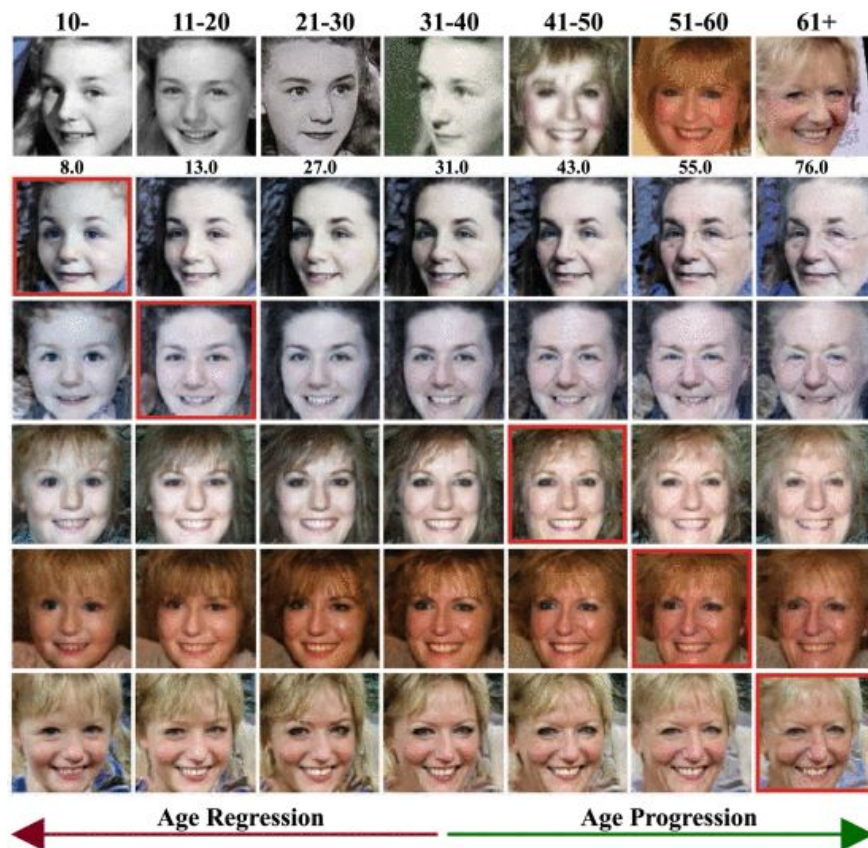


20 - 22 years

Left: Example application from course materials
Right: Image of missing child Ben Needham ([src](#))

Introduction – Challenges

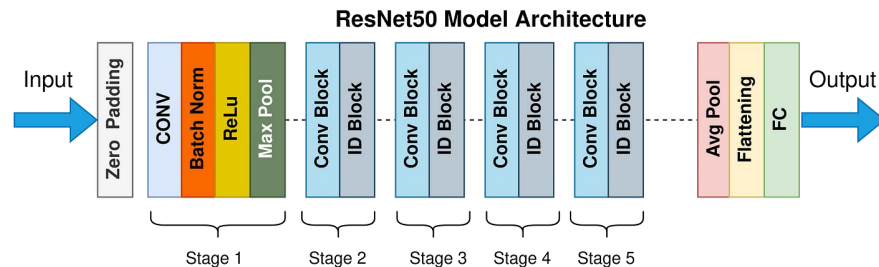
- Facial appearances change drastically after a long period of time
- How to extract features at the identity level and the aging level
- How to synthesize accurate, interpretable facial images



Right: Sample results of the MTLFace network ([src](#))

Background – General Face Recognition

- Goal:
 - Verify faces of the same identity regardless of variations in pose, age, expression, etc.
 - Distinguish between faces of different identities
- High accuracy from recent state-of-the-art models
 - Commonly use CNN to extract features (e.g. ResNet-100)
 - Extensive research on appropriate loss functions



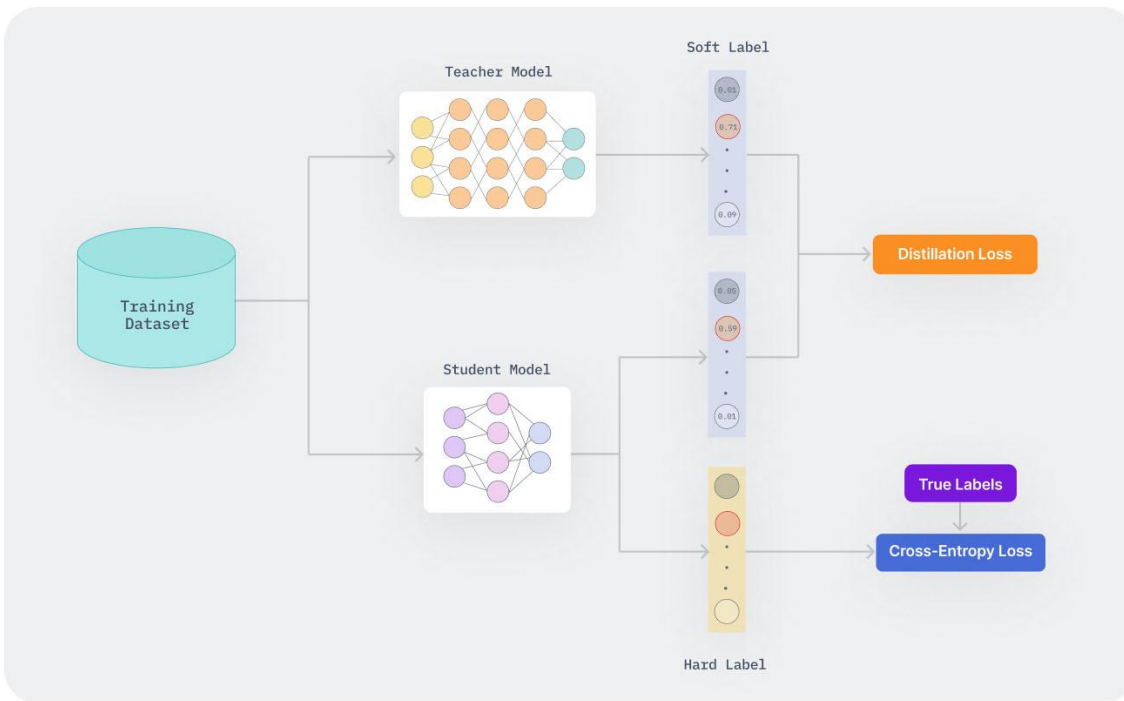
[src](#)

Method	LFW Acc.	AgeDB-30 Acc.
ArcFace	99.82%	98.15%
GroupFace	99.85%	98.28%
Partial-FC-ArcFace	99.83%	98.20%

[src](#)

Background – Knowledge Distillation

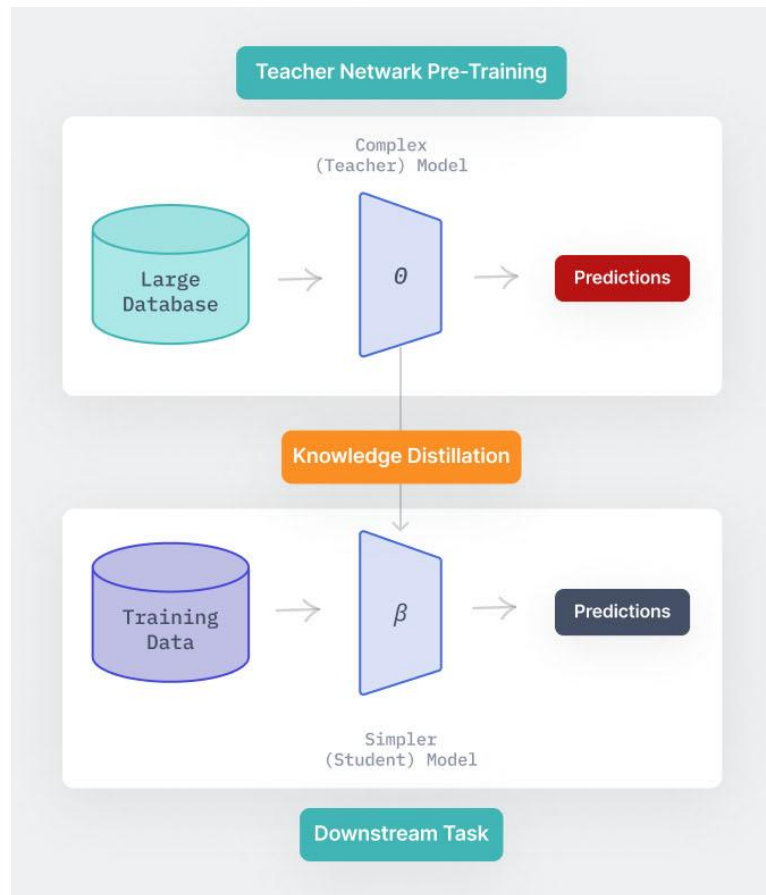
- KD uses both “soft” and “hard” targets to train the student
- Student not only able to generalize but generalize the same way the teacher does
- Advantages:
 - Better performance
 - Smaller dataset
 - Less computationally expensive to execute



Overview of the knowledge distillation training framework ([src](#))

Background – Approach 1

- Using a trained general face cognition (GFR) model with knowledge distillation (KD) to train a smaller model specific for performing age-invariant face recognition (AIFR)



Right: High-level view of knowledge distillation ([src](#))

Evolution of Age Invariant Face Recognition (AIFR)

2000s - Computer Vision Techniques:

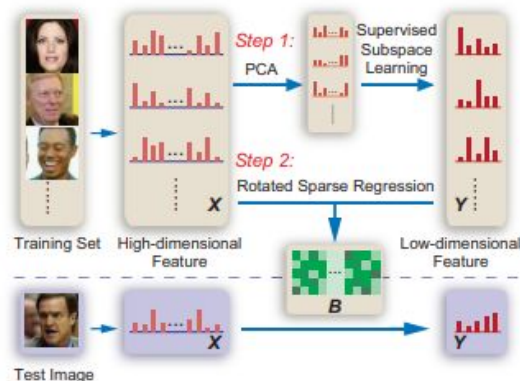
- Traditional methods relied on computer vision techniques like edge detection, template matching, and feature extraction.

	Approach	Face matcher	Database (#subjects, #images) in probe and gallery	Rank-1 identification, accuracy (%)	
				original image	after aging model
Ramanathan et al. (2006) [3]	Shape growth modeling up to age 18	PCA	Private database (109,109)	8.0	15.0
Lanitis et al. (2002) [4]	Build an aging function in terms of PCA coefficients of shape and texture	Mahalanobis distance, PCA	Private database (12,85)	57.0	68.5
Geng et al. (2007) [5]	Learn aging pattern on concatenated PCA coefficients of shape and texture across a series of ages	Mahalanobis distance, PCA	FG-NET ⁺ (10,10)	14.4	38.1
Wang et al. (2006) [6]	Build an aging function in terms of PCA coefficients of shape and texture	PCA	Private database (NA,2000)	52.0	63.0
Patterson et al. (2006) [7]	Build an aging function in terms of PCA coefficients of shape and texture	PCA	MORPH ⁺ (9,36)	11.0	33.0
Proposed method	Learn aging pattern based on PCA coefficients in separated 3D shape and texture given 2D database	FaceVACS	FG-NET ⁺⁺ (82,82)	26.4	37.4
			MORPH-Album ⁺⁺ (612,612)	57.8	66.4
			BROWNS (4,4) - probe (100,100) - gallery	15.6	28.1

Comparison of pre 2010 AFIR methods ([src](#))

2013 - High Dimensional LBP (Local Binary Patterns):

- Introduces a robust feature descriptor based on local texture patterns.
- Handles variations in facial appearance better than previous methods.

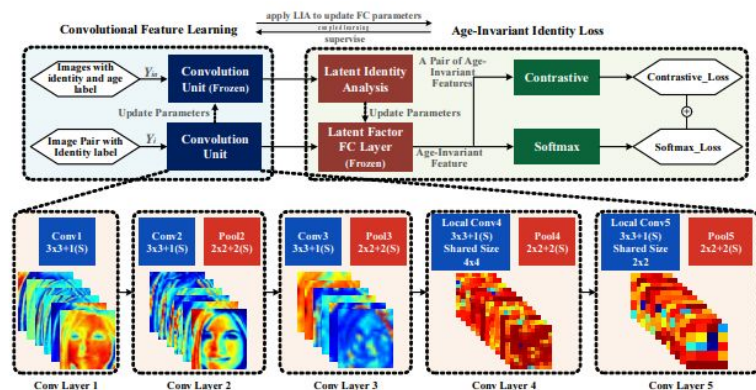


High-Dimensional LBP model ([src](#))

Introduction of Deep Learning in AIFR

2015 - MFM CNN (Multi-Feature Map CNN):

- Utilizes a convolutional neural network (CNN) architecture with multiple feature maps.
- Captures diverse facial features across different layers of the network.



MFM CNN model ([src](#))

2018 - AIM (Age-Invariant Model):

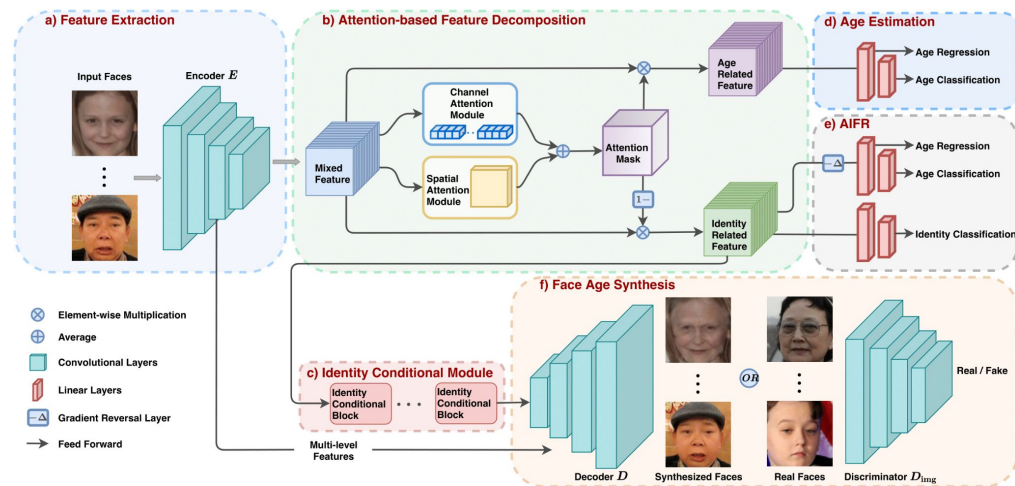
- Addresses the challenge of age variations in face recognition.
- Utilizes deep learning techniques to learn age-invariant features from facial images.



Synthesizing aged faces to learn features with AIM ([src](#))

Approach 2 - MTLFace (2021)

- Combines age-invariant face recognition (AIFR) and face age synthesis (FAS)
- Attention-based feature decomposition to separate identity-related from age-related features
- Uses synthesized faces to enhance recognition accuracy



Overview of the MTLFace methodology ([src](#))

MTLFace Performance Comparison

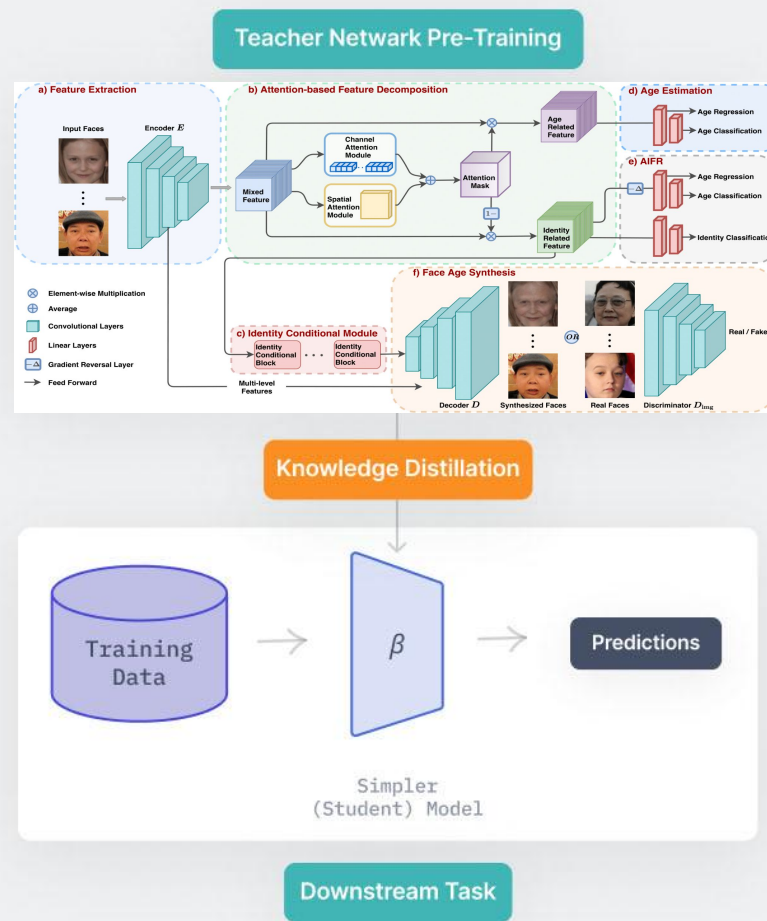
- MTLFace is our baseline in terms of performance, all of our work will be compared to this.
- It has 9-10% better prediction accuracy than other AIFR models.

Method	MORPH	FG-NET	CACD	ECAF
CAAE [15]	39.77/8.83/0.131±0.089	40.72/9.32/0.146±0.097	39.50/8.74/0.118±0.092	41.36/8.75/0.139±0.100
IPCGAN [13]	58.44/4.95/0.608±0.109	61.22/4.32/0.453±0.122	62.04/3.99/0.452±0.129	61.92/4.16/0.511±0.133
S2GAN [60]	53.85/5.47/0.369±0.105	60.26/4.37/0.293±0.111	60.19/4.31/0.270±0.110	62.35/3.98/0.297±0.117
MTLFace (ours)	67.07/2.98/0.652±0.091	71.33/3.10/0.648±0.094	71.94/2.55/0.630±0.096	72.14/2.28/0.620±0.096
w/o AIFR	51.79/5.49/0.219±0.095	48.60/7.68/0.235±0.093	50.69/6.50/0.197±0.095	52.31/6.14/0.217±0.103
w/o Multi-level ICMs	64.80/3.35/0.606±0.090	69.35/3.27/0.628±0.094	67.79/3.02/0.613±0.098	70.77/2.65/0.601±0.099
w/o ICM	62.60/3.86/0.605±0.072	65.46/3.65/0.612±0.080	62.55/3.87/0.591±0.099	67.34/3.16/0.584±0.088

Comparison of model performance ([src](#))

Initial Steps

- Goal is for approach 1 to have a higher accuracy than approach 2.
- Both models will be trained on two data sets.
- Adjustments will be made such as combining approach 1 and 2.



Right: Combining approaches 1 and 2.

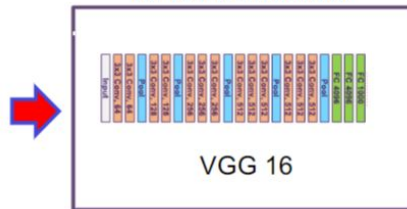
Methodology

- Datasets
- Feature extractor
- Loss function
- Metrics for evaluation

Create a dataset



Choose a deep learning network
for feature extractor



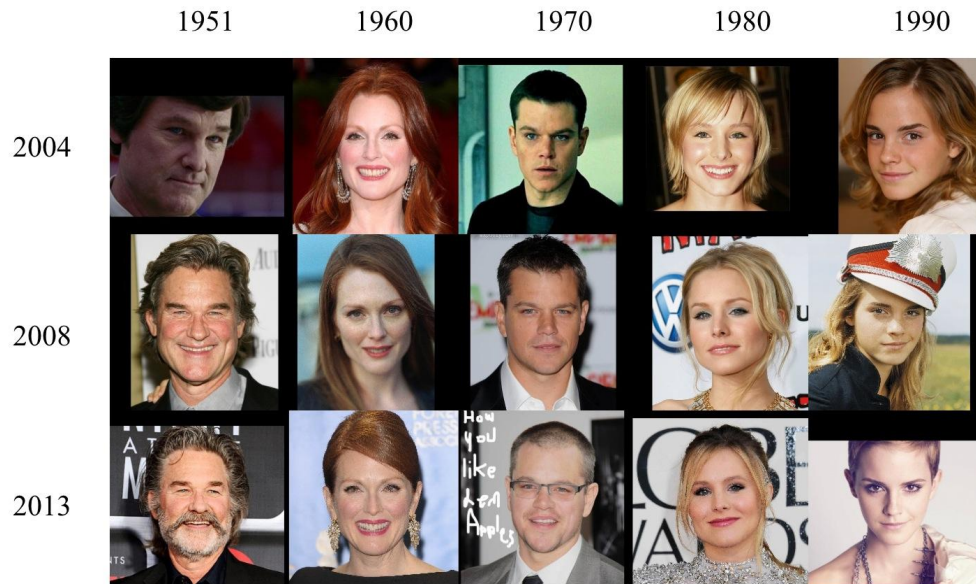
Use a proper loss function

Demonstration from course materials

Dataset - B3FD

Biometrically Filtered Famous Figure Dataset

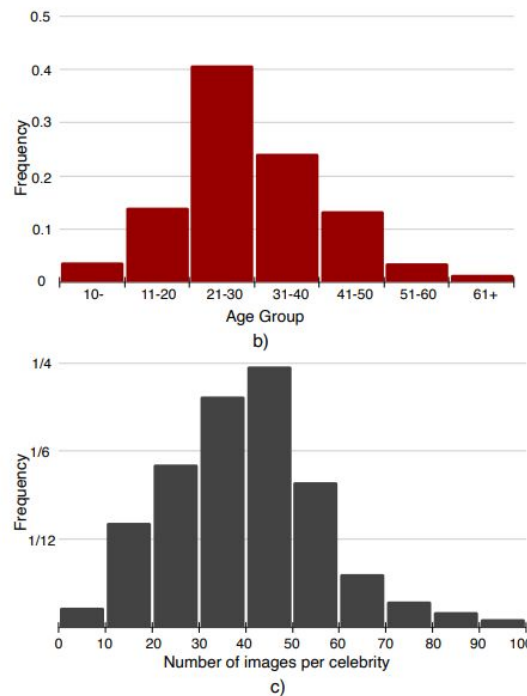
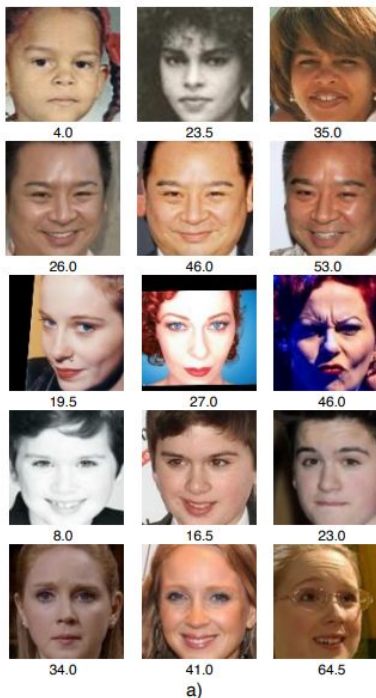
- 375,592 facial image samples with 53,759 unique subjects
- Web-scraped photos of same celebrities in different age periods
- The age labels are ranging from 0 to 100



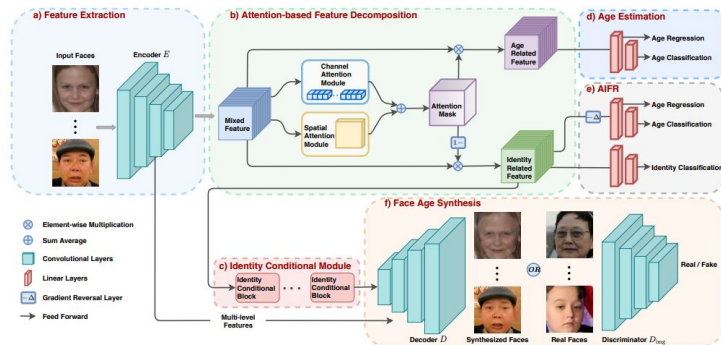
Dataset - LCAF

Large Cross-Age Face dataset

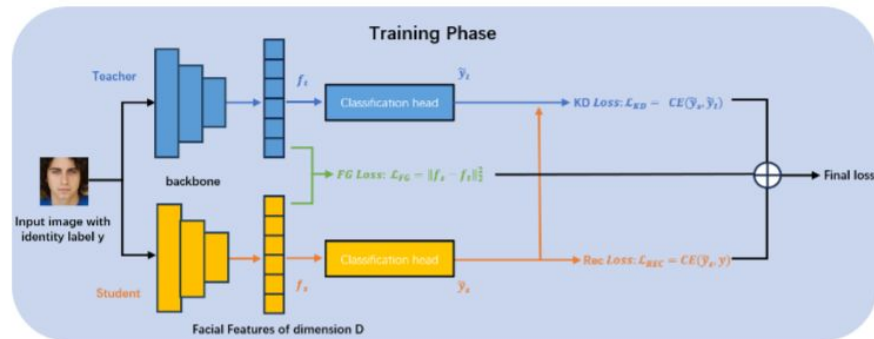
- 1.7 million faces of cross-age celebrities
- Balance at age and gender distributions



Feature extractors

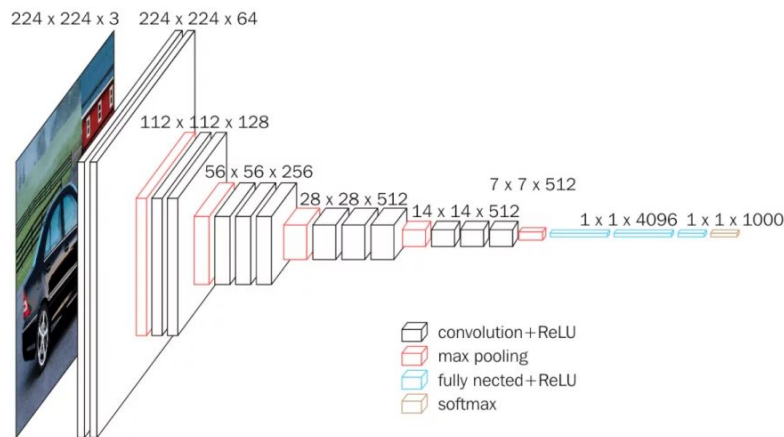


MTLFace

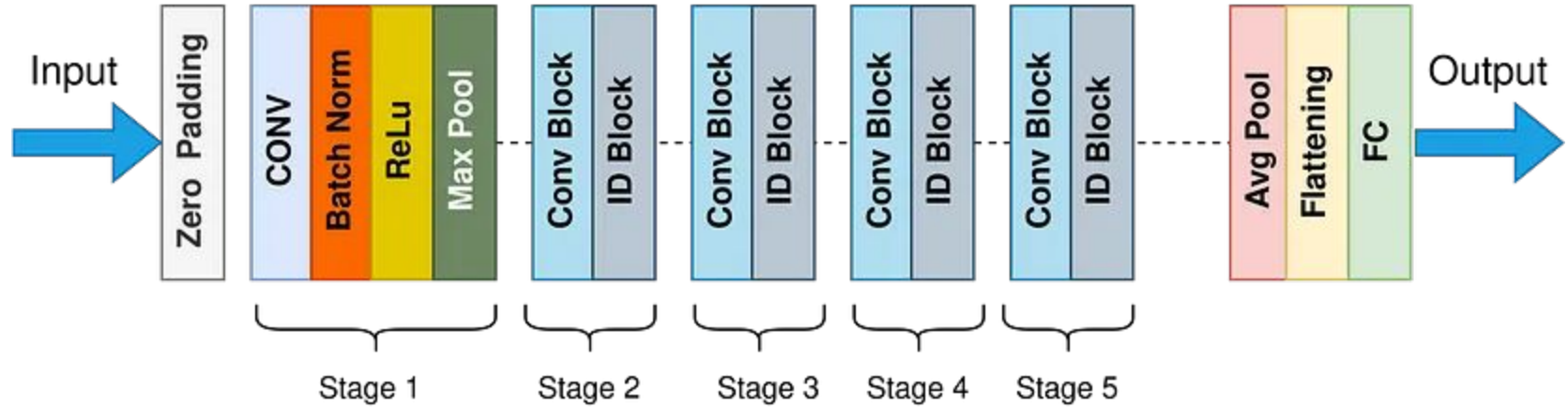


OursKD

Feature extractors - VGG 16



Feature extractors - ResNet 50



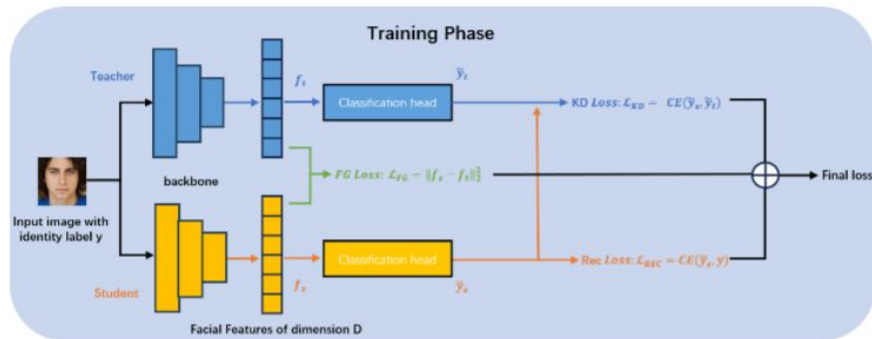
Loss function - OursKD

$$L_1 = ||f_t - f_s||_2^2$$

$$L_2 = \text{CE}(\hat{y}_t, \hat{y}_s)$$

$$L_3 = \text{CE}(y, \hat{y}_s)$$

$$L_{\text{total}} = L_1 + L_2 + L_3$$



Loss function - MTLFace

- Age estimation:

$$\ell_{AE}(\mathbf{X}_{age}) = \mathbb{E}_I [\ell_{MSE}(\text{DEX}(A(\mathbf{X}_{age})), y_{age}) + \ell_{CE}(A(\mathbf{X}_{age})\mathbf{W}, c_{age})],$$

- Age-invariant face recognition:

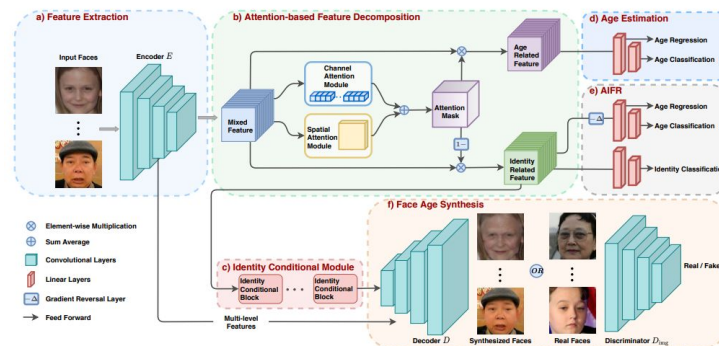
$$\mathcal{L}^{AIFR} = \ell_{\text{COSFACE}}(L(\mathbf{X}_{id}), y_{id}) + \lambda_{age}^{AIFR} \mathcal{L}_{AE}(\mathbf{X}_{age}) + \lambda_{id}^{AIFR} \mathcal{L}_{AE}(\text{GRL}(\mathbf{X}_{id}))$$

- Face age synthesis

$$\mathcal{L}^{FAS} = \lambda_{adv}^{FAS} \mathcal{L}_{adv}^{FAS} + \lambda_{id}^{FAS} \mathcal{L}_{id}^{FAS} + \lambda_{age}^{FAS} \mathcal{L}_{age}^{FAS},$$

$$\mathcal{L}_{adv}^{FAS} = \frac{1}{2} \mathbb{E}_I [D_{\text{img}}([I_t; C_t]) - 1]^2,$$

$$\begin{aligned} \mathbf{X}_{age}^t, \mathbf{X}_{id}^t &= \text{AFD}(E(I_t)) \\ \mathcal{L}_{age}^{FAS} &= \ell_{CE}(\mathbf{X}_{age}^t, t), \\ \mathcal{L}_{id}^{FAS} &= \mathbb{E}_{\mathbf{X}_s} \|\mathbf{X}_{id}^t - \mathbf{X}_{id}\|_F^2, \end{aligned}$$



Metrics - OursKD

- Feature map similarity

Cosine similarity between student's feature map and teacher's feature map

- Identity accuracy

Accuracy of correctly classifying the identity of individuals

Metrics - MTLFace

- Age accuracy

The proportion of predicted ages that fell within the target age groups

- Identity preservation

Cosine similarity between input and synthesized faces

Timeline

